

SRI VENKATESHA MANI NARAYANAN

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Professional Summary

Software Engineer with 2+ years of production experience across Full Stack Development, Cloud Native architectures and AI/ML applications. Proficient in Python, JavaScript and TypeScript with experience building production applications using FastAPI, Flask, Next.js, React.js, Node.js, and Express.js. Hands-on experience with Docker, Kubernetes, GCP, and AWS including EC2, Lambda, ECS, Fargate, S3, and SQS. Specializing in LLMs, RAG, AI agents, LangChain/LangGraph, MCP, and Vector Database.

Technical Skills

Languages: Python, JavaScript, TypeScript, Ruby, Java, SQL, HTML, CSS.

Frameworks & API: FastAPI, Flask, Node.js, Express.js, Next.js, React.js, Rails, REST APIs, GraphQL, WebSockets

AI / ML: AWS Bedrock, LangChain, LangGraph, Model Context Protocol (MCP), RAG, Vector Search, Agentic Workflows

Database & Messaging: PostgreSQL/pgvector, MySQL, MongoDB, Redis, Kafka, Snowflake.

Cloud & Infra: AWS(EC2, Lambda, ECS, Fargate, S3, SQS, RDS, EKS), Docker, Kubernetes, Terraform, Ansible

DevOps: CI/CD, GitHub Actions, Jenkins, CloudWatch, Prometheus, Grafana

Work Experience

Software Engineer | [Rattlesnake Ramble Organization](#) | Boulder, CO

Aug 2025 – Aug 2026

- Led a team migrating a **legacy PHP** application to a modern **PERN platform**, rebuilding **20+ features** and executing an **ETL migration to PostgreSQL** while architecting serverless workloads with **AWS Lambda and SQS**, driving **78% registration growth**.
- Resolved a critical **payment-state failure** by implementing **async PayPal webhooks with Amazon SQS and Lambda**, adding server-side verification, **idempotency, retries, and reconciliation** to maintain accurate payment state across **70+ registrations**.
- Migrated **2,500+ historical racer records spanning 20 years** from legacy **MySQL to PostgreSQL** through an **ETL pipeline**, transforming schemas, preserving **relational integrity**, and performing post-migration validation for production cutover.
- Deployed a serverless production architecture using **AWS Lambda with Amazon S3** for persistent media storage, implementing a **CI/CD pipeline with GitHub Actions** to automate builds and deployments and reduce production deployment time by **75%**.
- Developed an **LLM-powered admin assistant integrated into the React dashboard**, using **tool calling** to answer natural-language queries across registration and payment data, consolidating **7+ administrative workflows** into one conversational interface.

Software Engineer – AI & Full Stack | [Biocollate](#) | Boulder, CO

Aug 2024 – May 2025

- Owned end-to-end development of an **AI-powered LIMS/ELN**, containerizing **React and FastAPI** services and deploying them on **AWS ECS/Fargate** while integrating **RAG and agentic workflows** across **27 SBOL/LabOP APIs** serving **30+ lab researchers**.
- Built dependency-aware lifecycle management for **100+ biological components**, implementing **recursive traversal** to preserve component hierarchies, version history, and **RBAC**, reducing component lookup and **dependency-tracing time by 60%**.
- Engineered **LabOP-based ELN workflows** for shared protocol authoring and execution, implementing **versioned records, audit trails, and dual-signature verification** with independent witnesses, **reducing experiment documentation time by 50%**.
- Implemented **CatalystX**, a **RAG-powered AI assistant** using **LangChain**, LLM APIs, embeddings, and **PostgreSQL/pgvector** to semantically retrieve components and experiments, eliminating manual searches for **90% of routine researcher queries**.
- Developed **agentic workflows using LangGraph and LLM tool calling** across Benchtop Lab Assistant services, enabling **CatalystX** to trace component dependencies and affected experiments in **one natural-language request versus 5+ manual lookups**.

Graduate Teaching Assistant | [University of Colorado Boulder](#) | Boulder, CO

Aug 2024 - May 2025

- Mentored **100+ students in C++ and core programming concepts**, diagnosing complex implementation and debugging issues across object-oriented programming, recursion, pointers, memory, and data structures.
- Designed and graded **assignments and examinations** and led technical office hours, translating low-level programming concepts into systematic debugging strategies that strengthened students' problem-solving and code reasoning.

Software Development Intern | [ITJobxs](#) | Chennai, India

May 2024 – Aug 2024

- Delivered an end-to-end interview-experiences platform with **React and Express.js**, engineering **authentication, user verification, and MySQL-backed REST APIs** that enabled **500+ users** to securely publish and search interview experiences.
- Prevented **20–50 fake registrations per day** by implementing **O(1) hash-based email domain validation** and **Google reCAPTCHA** checks, intercepting bot traffic before account creation and MySQL insertion while strengthening signup security.
- Deployed the production application on **AWS EC2** with **Route 53** DNS routing and **CloudFront CDN** caching, configuring **Nginx as a reverse proxy** for the Node.js/Express backend and reducing asset load times by **35%**.

Software Development Intern | [Techfidelite Solutions](#) | India

May 2022 – Aug 2022

- Engineered secure **Python/Flask REST APIs** with **JWT-based authentication and authorization**, implementing token validation and protected request flows to enforce access control across backend services.
- Strengthened API security by implementing **IP blacklisting, whitelisting, request throttling, and validation middleware**, reducing exposure to abusive traffic while preserving reliable access for authenticated application clients.
- Built and refined **database-backed API workflows** while contributing through **GitHub, Jira, sprint planning, debugging, and technical documentation**, gaining production experience across the complete software development lifecycle.

Projects

PaperPilot | [GitHub](#)

- Built a **research-paper intelligence system**, orchestrating arXiv ingestion with **Apache Airflow**, Docling-based PDF parsing, chunking, embeddings, PostgreSQL metadata persistence, and **OpenSearch hybrid BM25/vector indexing** for retrieval.
- Engineered the **LangGraph agentic RAG workflow** with **guardrails, tool-based retrieval, document grading, query rewriting, Redis caching, and Langfuse tracing**, containerizing services with **Docker** and deploying on **AWS ECS/Fargate**.

Frameworkless RAG Engine | [GitHub](#)

- Built a **RAG pipeline from scratch without LangChain or LlamaIndex**, implementing PDF parsing, chunking, **384-dimensional embeddings**, **OpenSearch** vector indexing, similarity scoring, retrieval, and local generation with **Ollama**.
- Implemented **hybrid search, ranking, and retrieval directly in OpenSearch**, handling raw search responses, document metadata, relevance scores, and result merging while exposing retrieval behavior through an interactive **Streamlit** application.

RecipeRealm | [GitHub](#)

- Built RecipeRealm, a **full-stack MERN recipe-sharing platform** with **JWT-secured** user accounts, authentication, and social interactions, designing **REST APIs** and persistent application workflows across frontend and backend services.
- Containerized the application with **Docker** and deployed it on **AWS EC2** with **S3-backed image and video storage**, implementing automated testing with **Jest** and achieving **85% test coverage**.

AI-Generated Text Detection | [GitHub](#)

- Fine-tuned and compared **BERT, DistilBERT, and GPT-2** using **PyTorch** and **Hugging Face** across multi-source human and AI-generated text datasets, building preprocessing and model-training pipelines for binary text classification.
- Optimized **training configurations and hyperparameters** across transformer architectures and evaluated model performance on held-out data, producing a detection pipeline that achieved **82.7% evaluation accuracy**.

CERTIFICATIONS

IBM | [RAG and Agentic AI Professional Certificate](#)

Retrieval-Augmented Generation, Vector Databases, Multimodal Gen AI, Agentic AI, LangChain, LangGraph, CrewAI, BeeAI, MCP.

Anthropic | [Building with the Claude API](#) • [AI Fluency for Builders](#) • [Claude 101](#)

Claude API, Tool Calling, AI Agents, Prompt Engineering, LLM Evaluation, Responsible AI.

Education

MS in Computer Science | [University of Colorado Boulder](#) | 3.91/4.0 | *Boulder, CO* **2023 – 2025**

Relevant Coursework: Foundations of Software Engineering, Database Systems, Machine Learning, Natural Language Processing, Neural Networks & Deep Learning, Data Mining, User Centered Design & Development

B.Tech in Computer Science & Engineering | [SRM Institute of Science & Technology](#) | 9.29/10.0 | *Chennai, India* **2019 – 2023**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks